



Property Calculations and Design Examples

Isolation Bearing Design

rivtbok Example

Proj. 0004



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0.2-1 | Bearing Shape Factor

The stiffness of rubber in compression, when the loaded surfaces are prevented from slipping by bonding or by mechanical location, depends upon the shape factor S defined as the ratio of one loaded area to the total force-free area (see Fig. 1). **[0.2.1]**

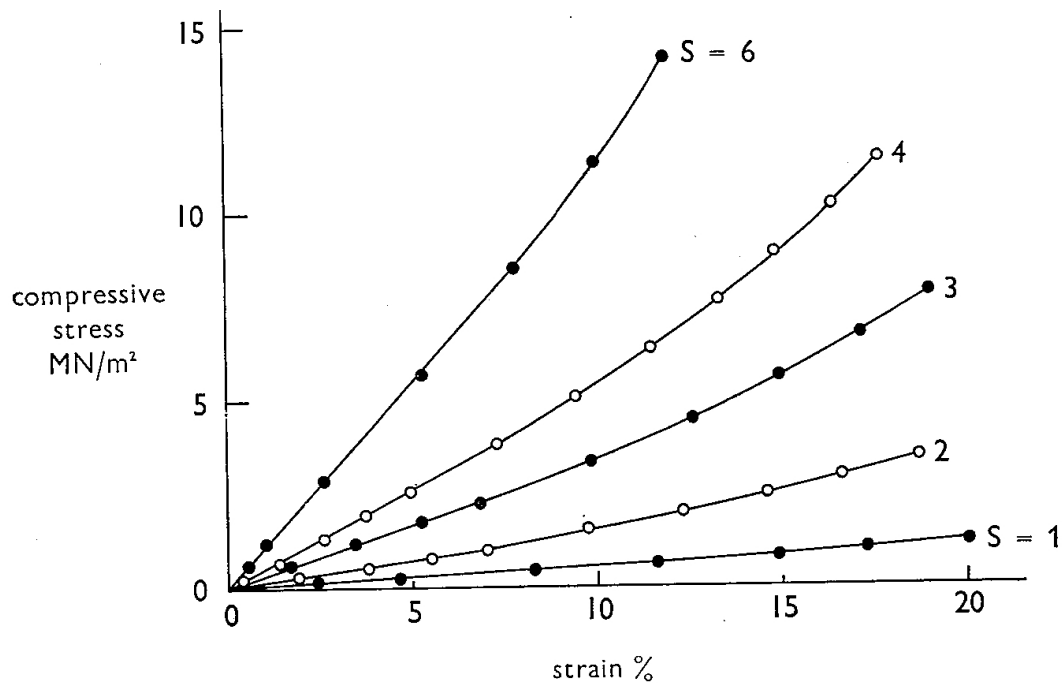


Fig. 1 - Shape Factor

Effect of shape factor: Experimental stress-strain curves for 6.3 mm thick disks of rubber (47 IRHD) in compression. The shape factor is shown alongside each curve; the diameter in mm is 25.4 times the shape factor.

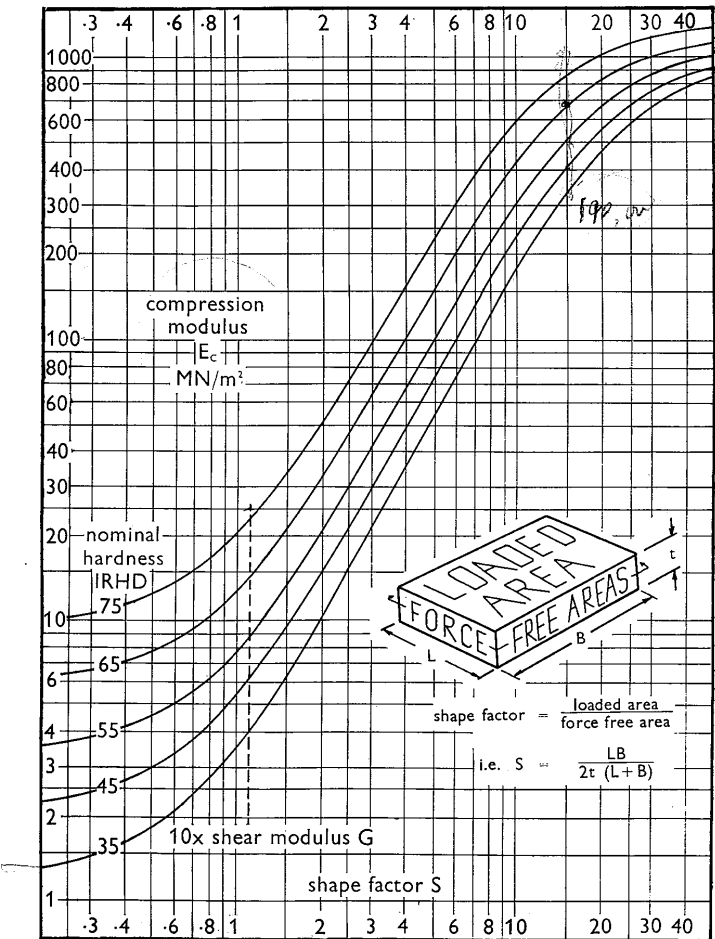


Fig. 28.—Variation of compression modulus E_c with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above).

* Although deformation due to bulk compression can normally be ignored, it can cause a noticeable reduction in E_c when the ratio E_c/E_∞ exceeds about 10%. (E_∞ is the modulus of bulk compression, from Table 3.) To allow for this reduction, use a modified compression modulus obtained by dividing E_c by $1+(E_c/E_\infty)$. If Wood and Martin's value for bulk modulus is preferred (see footnote to Table 2) then the E_∞ value in Table 3 should be doubled.

Fig. 2 - Compression Modulus

Variation of compression modulus E_c with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above). Although deformation due to bulk compression can normally be ignored, it can cause a noticeable reduction in E_c when the ratio E_c/E_∞ exceeds about 10%. (E_∞ is the modulus of bulk compression) To allow for this reduction, use a modified compression modulus obtained by dividing E_c by $1+(E_c/E_\infty)$. If Wood and Martin's value for bulk modulus is preferred then the E_∞ value in Table 3 should be doubled.

Table 1: IRHD hardness scale (MN/m²) (stored: t002-1.csv)

Hardness	Young's mod E_0	Shear mod G	k	Bulk mod
30	0.92	0.30	0.93	1000
35	1.18	0.37	0.89	1000
40	1.50	0.45	0.85	1000



Hardness	Young's mod E0	Shear mod G	k	Bulk mod
45	1.80	0.54	0.80	1000
50	2.20	0.64	0.73	1030
55	3.25	0.81	0.64	1090
60	4.45	1.06	0.57	1150
65	5.85	1.37	0.54	1210
70	9.40	2.22	0.52	1330

Table 2: Shore A hardness scale (lbf/in²) (stored: t002-2.csv)

Hardness	Young's mod E0	Shear mod G	k	Bulk mod
35	168	53	0.89	142 000
45	256	76	0.80	142 000
55	460	115	0.64	154 000
65	830	195	0.54	171 000
75	1340	317	0.52	189 000

Notes

- k is used in the calculation of compression characteristics.
- The majority of springs are in the hardness range 40-60 IRHD.
- Average design limits: 15% compression, 50% shear.
- Theoretically, with a Poisson's ratio of 1/2, E0 should equal 3G. This is so for soft gum rubbers, but for harder rubbers containing a fair proportion of non-rubber constituents, thixotropic and other effects increase E0 to about 4G.
- Based on experiments on natural rubber spring vulcanizates containing (above 48 IRHD) SRF black as filler. Note that hardness is subject to an uncertainty of about +/-2 degrees.

0.2-2 | Block Compression Stiffness**Table 3:** Shape Factor Parameters

variable	value	[value]	description
t1	0.25	0.25	thickness
L1	24	24	length
B1	12	12	breadth

For a rectangular section (length = L and breadth = B):

Eq. 1: shape factor

$$S_1 = \frac{B_1 \cdot L_1}{t_1}$$



$$2 \cdot t_1 \cdot (B_1 + L_1)$$

$$S_1 = 16.0 \quad [S_1] = 16.0 \quad | \text{ shape factor}$$

B_1	L_1	t_1
12.0	24.0	0.25
breadth	length	thickness

For a block of square section (i.e. $L = B$) or circular section (diameter = L):

Eq. 2: shape factor

$$S_2 = \frac{L_1}{4 \cdot t_1}$$

$$S_2 = 24.0 \quad [S_2] = 24.0 \quad | \text{ shape factor}$$

L_1	t_1
24.0	0.25
length	thickness

The compression modulus E_c depends upon the shape factor S (for derivation see Section 5).

Table 4: E_c parameters

variable	value	[value]	description
E_0	100.00 p_si	0.69 MPA	compression modulus
k_1	0.7	0.7	adjustment factor
G_1	0.50 k_si	3.45 MPA	shear modulus

Eq. 3: compression modulus

$$E_c = E_0 \cdot \left(2 \cdot S_1 \cdot k_1 + 1 \right)$$

$$E_c = 35.94 \text{ k_si} \quad [E_c] = 247.80 \text{ MPA} \quad | \text{ compression modulus}$$

S_1	k_1	E_0
16.0	0.7	100.00 p_si
shape factor	adjustment factor	compression modulus



When $S > 3$ it may be more convenient to use:

Eq. 4: compression stiffness - modified

$$E_{lc} = 5 \cdot G_1 \cdot S_1^2$$

$E_{lc} = 640.00 \text{ k_si}$ $[E_{lc}] = 4412.64 \text{ MPA}$ | compression stiffness - modified

G_1	S_1
0.50 k_si	16.0
shear modulus	shape factor

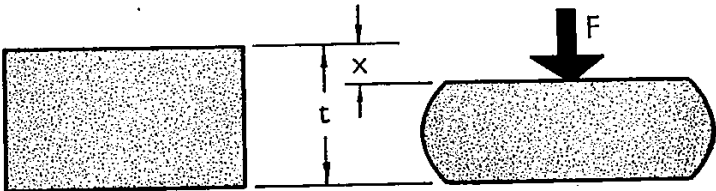


Fig. 29.—Rubber block in compression.

Fig. 3 - Compression Stiffness Kc

The diagram shows a rectangular rubber block of thickness t , deflecting by x under a compressive load F applied to a cylindrical rubber sample.

The compression stiffness K_c is given by:

Eq.5:

$$K_c = \frac{F}{x}$$

Table 5: Stiffness parameters

variable	value	[value]	description
A1	20.00 in2	12903.20 mm2	cross-sectional area
t1	0.25 inch	6.35 mm	thickness
x1	1.00 inch	25.40 mm	deflection

Eq. 6: compression stiffness

$$K_c = \frac{A_1 \cdot E_c}{t}$$



t_1		
$K_c = 2875.20 \text{ k_in} \quad [K_c] = 5035.25 \text{ kN_cm} \quad \text{ compression stiffness}$		
E_c	A_1	t_1
35940.00 p_si	20.00 in2	0.25 inch
compression modulus	cross-sectional area	thickness

If $S > \mu/2$, where S is the shape factor and μ the coefficient of friction, slip may occur. In such cases, the rubber should be bonded to the rigid

The load (F)—deflexion (x) curve of rubber in compression is non-linear. With no slip it has the approximate form

Eq.7:

$$F = A \cdot E_c \cdot e \cdot (e + 1)$$

where e , the compressive strain, equals x/t . The non-linearity is usually ignored for strains up to about 10%.
There is as yet no method of calculating that a block will be stable but experience has shown that provided the thickness is less than one-quarter of the least plan dimension there should be no instability.

0.2-3 | Strip Compression Stiffness

When a long strip of rubber is compressed the strain in the direction of its length will be negligible.

Shape factor

Eq.8:

$$SF = \frac{b}{2 \cdot t}$$

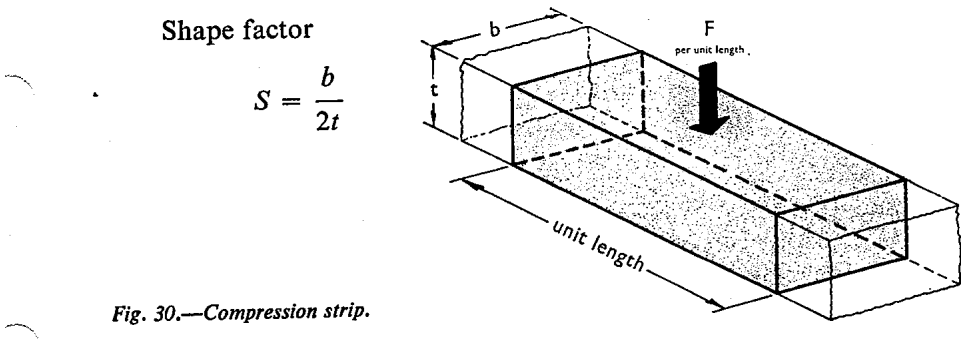


Fig. 4 - Compression Strip



Diagram shows a long rectangular strip of rubber of width b , thickness t , and unit length, compressed by a load F applied per unit length.

Compression modulus E_c

Eq.9:

$$E_c = \frac{4 \cdot E_0 \cdot \left(\frac{2}{SF \cdot k + 1} \right)}{3}$$

The compression stiffness per unit length, K_c , is given by

Eq.10:

$$K_c = \frac{F}{x}$$

Eq.11:

$$K_c = \frac{E_c \cdot b}{t}$$

Eq.12:

$$K_c = \frac{4 \cdot E_0 \cdot b \cdot \left(\frac{2}{SF \cdot k + 1} \right)}{3 \cdot t}$$

F = load per unit length

E_c = compression modulus (corrected, if necessary, for the effect of bulk compression)

E_0 = Young's modulus (from Table 3)

b = width of strip

t = thickness of strip

x = deflexion

k = a numerical factor (from Table 3)

SF = shape factor

The load per unit length (F)—deflexion (x) curve for a compressed strip is non-linear. It has the approximate form

Eq.13:



$$F = E_c \cdot b \cdot e \cdot \left(\frac{3 \cdot e}{2} + 1 \right)$$

where e , the compressive strain, equals x/t . As in the case of blocks, the non-linearity is usually ignored for strains up to about 10%.

0.2-4 | Derivation of Compression Characteristics

Approximate load-deformation relations are derived below for small compressions of rubber blocks, between rigid plates to which they adhere or are bonded. The total displacements are considered to arise from the superposition of simple displacements, namely, (1) the pure homogeneous deformation defined by the displacement of one rigid bonding plate towards the other, and (2) the subsequent displacements necessary to cause points in the planes of the bonded surfaces to be restored to their original positions in these planes.

‘[0.2.2]’_

Two extreme cases are considered, infinitely long rectangular blocks, the infinite dimension being at right angles to the direction of compression, and circular disks, loaded axially.

Rectangular Blocks of Infinite Length **_bold**

The pure homogeneous deformation consists of a compressive strain e in the vertical direction, zero strain in the length direction, and an expansion in the width direction given, since the rubber is virtually incompressible in volume, by e also. The rubber is placed in a state of pure shear and the force F_1 which has to be applied to the bonded surfaces to maintain such a deformation is given by

Eq.14:

$$F_1 = \frac{4 \cdot E_0 \cdot b \cdot e}{3}$$

per unit length, where E_0 is the Young's modulus of the rubber and b is the width of the block.

The second displacement system is that causing points in the planes of the bonded surfaces to be restored to their original positions in these planes. In order to calculate the corresponding system of forces which has to be applied to the bonded surfaces, it is necessary to make the simplifying assumption that horizontal planes remain plane during the deformation, as seems probable when the width b is much greater than the thickness t . Vertical sections consequently take up parabolic forms, as represented diagrammatically in Fig. 41.

The maximum outward displacement of any originally vertical section is dictated by the requirement that the volume contained between it and the central plane in the undeformed state shall equal the volume contained between it and the central plane when it takes up the parabolic form and the bonded surfaces approach by an amount $\delta (= e \cdot t)$. ‘[0.2.3]’_

For a plane at a distance x from the central vertical plane, considering unit length, we have

Eq.15:

$$t \cdot x = \frac{2 \cdot t \cdot u}{3} + x \cdot (-\delta + t)$$



where $2ut/3$ is the approximate volume contained between the parabolic front of maximum displacement u and the vertical plane at x , when the compression δ is small. Hence,

Eq.16:

$$u = \frac{3 \cdot \delta \cdot x}{2 \cdot t}$$

In an elementary section of width dx the small displacement u may be maintained by the action of an excess hydrostatic pressure dp acting on one curved face, where dp is given by the classical theory of elasticity in the form

Eq.17:

$$dp = dx \cdot \frac{-8 \cdot E_0 \cdot u}{3 \cdot t^2}$$

Since the curved surface at $x = b/2$ is force free, p may be obtained by integrating both sides of equation (2) between the limits $x = x$ and $x = b/2$, yielding

Eq.18:

$$p = \frac{2 \cdot E_0 \cdot \delta \cdot \left(\frac{b^2}{4} - x^2 \right)}{3 \cdot t}$$

The pressure p acts on the rigid bonded plates also. The corresponding force F_2 acting on each of these surfaces may be obtained by integrating such terms as $p \cdot dx$ between the limits $x = b/2$ and $x = -b/2$, to give

Eq.19:

$$F_2 = \frac{E_0 \cdot b \cdot \delta}{3 \cdot t}$$

The apparent value of Young's modulus for the rubber block is given by

Eq.20:



$$\square = F \cdot \frac{1}{b} \cdot \frac{1}{e}$$

where F is the total force which has to be applied to the bonded surfaces per unit length to maintain the deformation, and is given by $F_1 + F_2$ where F_1 and F_2 are given by the equations. Hence,

Eq.21:

$$\square = \frac{E_0 \cdot b}{2} + \frac{4 \cdot E_0}{3 \cdot t}$$

where the shape factor S , the ratio of one loaded surface to the force-free surface, is equal to $b/(2 \cdot t)$

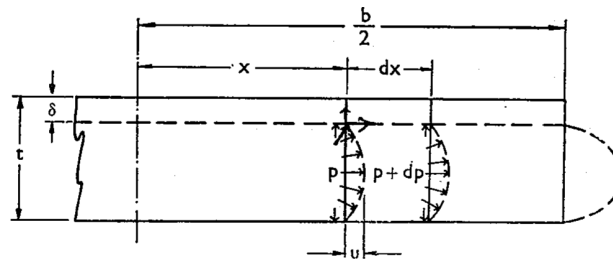


Fig. 5 - Initial and Deformed States

Initial and deformed states for an infinitely long block subjected to a small compression.

Circular Disks

The apparent Young's modulus may be derived similarly for a circular disk. The force F_1 which must be applied to maintain the first deformation, a simple compression of amount e , is given by '[0.2.4]'

Eq.22:

$$F_1 = E_0 \cdot a^2 \cdot e \cdot \pi$$

where a is the radius of the disk.

The force F_2 which has to be applied to maintain the second displacement system may be calculated by a similar procedure to that used in the previous example. Initially, vertical cylindrical surfaces are assumed to take up parabolic forms such that the volume contained within them is unchanged. When the compressive displacement δ is small, and the radius r is much greater than the maximum radial displacement u , then

Eq.23:



$$u = \frac{3 \cdot \delta \cdot r}{4 \cdot t}$$

where t is the thickness of the disk. The pressure p acting at a distance r from the centre is found to be

Eq.24:

$$p = \frac{E_0 \cdot \delta \cdot \left(\frac{2}{a^2} - \frac{2}{r^2} \right)}{3t}$$

and hence the force F_2 is obtained as

Eq.25:

$$F_2 = \frac{4 E_0 \cdot a \cdot \delta \cdot \pi}{2 \cdot t^3}$$

An analogous relation has been derived by Dienes and Klemm for the rate of approach of two parallel circular plates separated by a viscous liquid.

The apparent Young's modulus E for a circular disc is obtained finally in the form

Eq.26:

$$E = E_0 \cdot \left(\frac{\frac{2}{a^2}}{\frac{2}{2 \cdot t^2}} + 1 \right)$$

where S , the ratio of one loaded surface to the force-free surface, is equal to $a/(2 \cdot t)$.

As the thickness of the block is increased, the terms representing the contribution F_2 to the total force decrease, so that when the condition on which the calculation for F_2 is based – that the width or radius is much greater than the thickness – becomes inapplicable. The contribution F_2 to the total force becomes relatively small. The equations may therefore be expected to apply over a considerable range of thicknesses.

NOTE: In practice it is necessary to multiply the S^2 term by the factor k , given in Table 1. '[0.2.5]'



0.3-1 | Rubber Elastic Properties

Table 1: IRHD hardness scale (MN/m²)

Hardness IRHD +/-2	Young's modulus E, MN/m ²	Shear modulus G, MN/m ²	k	Bulk modulus B(inf), MN/m ²
30	0.92	0.30	0.93	1000
35	1.18	0.37	0.89	1000
40	1.50	0.45	0.85	1000
45	1.80	0.54	0.80	1000
50	2.20	0.64	0.73	1030
55	3.25	0.81	0.64	1090
60	4.45	1.06	0.57	1150
65	5.85	1.37	0.54	1210
70	7.35	1.73	0.53	1270
75	9.40	2.22	0.52	1330

Table 2: Shore A hardness scale (lbf/in²)

Shore A (approx.)	Young's modulus lbf/in ²	Shear modulus lbf/in ²	k	Bulk modulus lbf/in ²
35	168	53	0.89	142 000
45	256	76	0.80	142 000
55	460	115	0.64	154 000
65	830	195	0.54	171 000
75	1340	317	0.52	189 000

Note

1. The majority of springs are in the hardness range 40-60 IRHD.
2. Theoretically, with a Poisson's ratio of 1/2, B should equal 3G. This is true for soft gum rubbers. For harder rubbers containing a fair proportion of non-rubber constituents, thixotropic and other effects increase 3G to about 4G.

0.3-2 | Shear Stiffness Example

Table 3: Bearing Specifications

variable	value	[value]	description
G_1	58.00 p_si	0.40 MPA	shear modulus
K_1	200000.00 p_si	1378.95 MPA	bulk modulus
rn1	51	51	number of rubber layers
rdia	39.50 inch	100.33 cm	bearing diameter
rthk	0.40 inch	1.02 cm	layer thickness



variable	value	[value]	description
sdia	38.00 inch	96.52 cm	shim diameter
sthk	0.09 inch	0.23 cm	11 guage shim thickness
cpi	3.1418	3.1418	contant pi

Bearing Stiffness for Circular Bearing

Eq. 1: rubber height

$$rht = rnl \cdot rthk$$

$$rht = 20.40 \text{ inch} \quad [rht] = 51.82 \text{ cm} \quad | \text{ rubber height}$$

rnl	rthk
51	0.40 inch
number of rubber layers	layer thickness
-	-

Eq. 2: bearing height

$$bht = rht + sthk \cdot (rnl - 1)$$

$$bht = 24.95 \text{ inch} \quad [bht] = 63.37 \text{ cm} \quad | \text{ bearing height}$$

rht	rnl	sthk
20.40 inch	51	0.09 inch
rubber height	number of rubber layers	11 guage shim thickness
-	-	-

Eq. 3: shear stiffness

$$K_{s1} = \frac{G_1 \cdot cpi \cdot rdia^2}{4 \cdot rht}$$

$$K_{s1} = 3.48 \text{ k_in} \quad [K_{s1}] = 6.10 \text{ kN_cm} \quad | \text{ shear stiffness}$$

G ₁	cpi	rdia	rht
58.00 p_si	3.1418	39.50 inch	20.40 inch



<u>shear modulus</u>	<u>contant pi</u>	<u>bearing diameter</u>	<u>rubber height</u>
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Shape Factor for Circular Bearing

Eq. 4: shape factor 1

$$sh_{1 \ 1} = \frac{sdia}{4 \cdot rthk}$$

$$sh_{1 \ 1} = 24 \quad [sh_{1 \ 1}] = 24 \quad | \text{ shape factor 1}$$

<u>sdia</u>	<u>rthk</u>
38 inch	0 inch
<u>shim diameter</u>	<u>layer thickness</u>

Eq. 5: shape factor 2

$$sh_{2 \ 1} = \frac{bht}{rdia}$$

$$sh_{2 \ 1} = 1 \quad [sh_{2 \ 1}] = 1 \quad | \text{ shape factor 2}$$

<u>bht</u>	<u>rdia</u>
25 inch	40 inch
<u>bearing height</u>	<u>bearing diameter</u>

0.3-3 | shear

Natural Frequency

The natural frequency (n_f) in Hz of a mounted body on a spring is

$n_f = 5 / \text{sqrt}(x)$

where x is the effective deflexion of the spring in centimetres. The effective deflexion is affected (and usually decreases) by the amplitude effect and by non-linear springs (Fig. 16).

[Figure: fig16 _static_load_deformation_curve.png] Fig. 16.-OAB is a ‘static’ load deformation curve. Its exact position and shape will depend on the amount of structure breakdown. The static stiffness at A is the slope of the tangent AC. When a dynamic, small amplitude deformation ED is superimposed on the static deformation at A its



stiffness is the slope of AF. Thus the effective dynamic deformation FG is smaller than CG or OG, and the natural frequency will be higher than that predicted from static behaviour. C and F are generally closer to O than illustrated, especially with gum and lightly-filled natural rubber vulcanizates.

[Figure: fig17_transmissibility_graph.png] Fig. 17.-The transmissibility of a natural rubber containing 40 parts by weight of carbon black. Temperature 19 degrees C. Experimental data of Snowdon, J. C., Brit. J. Appl. Phys, 1958, 9, 461-9. The full line is the theoretical relationship for no damping.

Transmissibility

Transmissibility (T) is the ratio of the amplitude of the vibration on the 'protected' side of the spring to that of the disturbing vibration, or when the spring is on a rigid base, the ratio of the corresponding forces. T depends on the ratio of the disturbing frequency n to the natural frequency n_f of the mounted system (see above).

Under normal operating conditions when $n > 3 \cdot n_f$ natural rubber gives attenuation comparable with materials having no damping (Fig. 17). At resonance, when $n = n_f$, the small amount of damping in natural rubber prevents the peak value of T becoming excessive.

Stiffness Characteristics

A mathematical theory of rubber-like elasticity has been developed for elastic deformations of up to several hundred per cent. The theory applies strictly only to ideal rubbers, which are completely reversible, incompressible and isotropic, but in practice actual rubber springs have been shown to conform quite well. The behaviour of certain kinds of rubber spring may therefore be calculated for deformations of any practicable amount.

However, when the deformation is complex, an exact solution may be unattainable. In some of these cases approximate relationships can be derived from the classical theory of elasticity. This theory, which is the basis of standard engineering practice, only applies to materials in which the strains are very small (a few per cent), but the errors introduced by extrapolation to strains of the order of 10-20% are not excessive. In the stiffness equations which follow, terms beyond the first have only been given where their contribution is likely to exceed the variability of the elastic constants. Calculated stiffnesses should be within $\pm 15\%$ of the actual stiffness in most cases; small adjustments can be obtained by slight changes of rubber hardness.

In general, stiffness is particular to a given direction, the stiffnesses in other directions may sometimes be an order of magnitude different as, for example, in bridge bearings. By the suitable selection and location of two or more units the stiffness of a composite spring in three different directions can be varied independently. The design of such springs, of which the inclined shear mounting is a typical example, requires only a knowledge of mechanics and the principal stiffnesses of the units.

SHEAR MOUNTING

[Figure: fig18_shear_mounting.png] Fig. 18. Shear mounting.



0.4-1 | Equivalent Viscous Damping

Equivalent viscous damping is the simplest form of damping for analysis since the governing differential equation of motion is linear and hence amenable to analytical solution. This section determines the damping coefficient for viscous damping so that it is equivalent in some sense to the combined effect of all damping mechanisms present in the actual structure. [0.4.1]

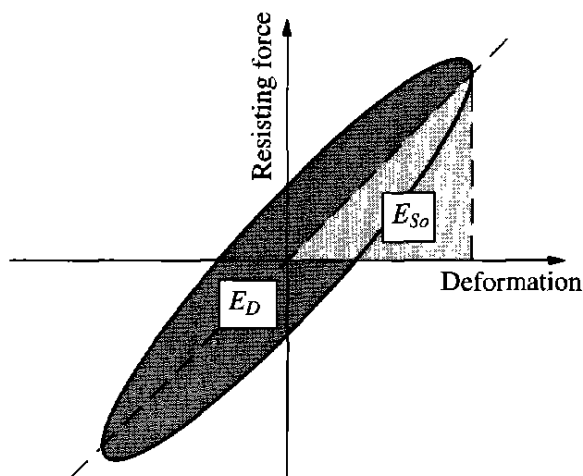


Fig. 1 - Ideal Viscous Damping Model

The simplest definition of equivalent viscous damping is based on the measured response of a system to harmonic force at exciting frequency ω equal to the natural frequency ω_n of the system. The damping ratio ζ_{eq} is calculated from using measured values of u_o and $(u_{st})_o$. This is the equivalent viscous damping since it accounts for all the energy-dissipating mechanisms that existed in the experiments.

0.4-2 | Measured Viscous Damping

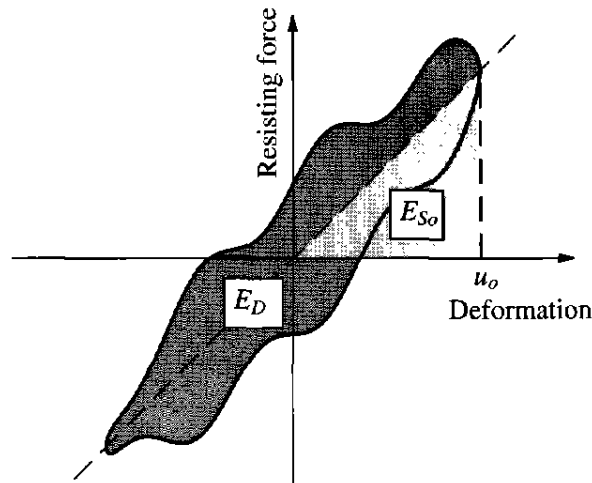


Fig. 2 - Measured Viscous Damping Model

The most common method for defining equivalent viscous damping is to equate the energy dissipated in a vibration cycle of the actual structure and an equivalent viscous system. For an actual structure the force-displacement relation obtained from an experiment under cyclic loading with displacement amplitude u_o is determined; such a relation of arbitrary shape is shown schematically in Fig. 2. The energy dissipated in the actual structure is given by the area E_D enclosed by the hysteresis loop. Equating this to the energy dissipated in viscous damping given by Eq.1 leads to

Eq.1

$$4 \pi \zeta_{eq} \frac{\omega}{\omega_n} E_{So} = E_D \quad \text{or, equivalently,}$$

Eq.2:

$$\zeta_{eq} = \frac{1}{4 \pi} \cdot \frac{1}{\left(\frac{\omega}{\omega_n} \right)} \cdot E_D \cdot \frac{1}{E_{So}}$$

where the strain energy, $E_{So} = ku_o^2/2$, is calculated from the stiffness k determined by experiment.

The experiment leading to the force-deformation curve of Fig. 3.9.1 and hence E_D should be conducted at $\omega = \omega_n$, where the response of the system is most sensitive to damping. Thus Eq. specializes to

Eq.3:

$$\zeta_{eq} = \frac{1}{4 \pi} \cdot E_D \cdot \frac{1}{E_{So}}$$

The damping ratio ζ_{eq} determined from a test at $\omega = \omega_n$ would not be correct at any other exciting frequency, but it would be a satisfactory approximation.

It is widely accepted that this procedure can be extended to model the damping in systems with many degrees of freedom. An equivalent viscous damping ratio is assigned to each natural vibration mode of the system in such a way that the energy dissipated in viscous damping matches the actual energy dissipated in the system when the system vibrates in that mode at its natural frequency.

In this book the concept of equivalent viscous damping is restricted to systems vibrating at amplitudes within the linearly elastic limit of the overall structure. The energy dissipated in inelastic deformations of the structure have also been modeled as equivalent viscous damping in some research studies. This idealization is generally not satisfactory, however, for the large inelastic deformations of structures expected during strong earthquakes. We shall account for these inelastic deformations and the associated energy dissipation by nonlinear force-deformation relations.

[0.4.1]

Anil K. Chopra, Dynamics of Structures: Theory and Applications to Earthquake Engineering. Englewood Cliffs, NJ, USA: Prentice Hall, 1995.



0.5-1 | Rate Independent Damping

Experiments on structural metals indicate that the energy dissipated internally in cyclic straining of the material is essentially independent of the cyclic frequency. Similarly, forced vibration tests on structures indicate that the equivalent viscous damping ratio is roughly the same for all natural modes and frequencies. Thus we refer to this type of damping as rate-independent linear damping. Other terms used for this mechanism of internal damping are structural damping, solid damping, and hysteretic damping. We prefer not to use these terms because the first two are not especially meaningful and the third is ambiguous because hysteresis is a characteristic of all materials or structural systems that dissipate energy. [0.5.1]

Rate-independent damping is associated with static hysteresis due to plastic strain, localized plastic deformation, crystal plasticity, and plastic flow in a range of stresses within the apparent elastic limit. On the microscopic scale the inhomogeneity of stress distribution within crystals and stress concentration at crystal boundary intersections produce local stress high enough to cause local plastic strain even though the average “global” stress may be well below the elastic limit. This damping mechanism does not include the energy dissipation in global plastic deformations, which, as mentioned earlier, is handled by a nonlinear relationship between force f and deformation u .

The simplest device that can be used to represent rate-independent linear damping is to assume that the damping force is proportional to velocity and inversely proportional to frequency:

Eq.1:

$$f_D = \frac{k \cdot u' \cdot \eta}{\omega}$$

where k is the stiffness of the structure and η is a damping coefficient. The energy dissipated by this type of damping in a cycle of vibration at frequency ω is independent of ω (Fig. 3.10.1). It is given by Eq. (3.8.1) with c replaced by $\eta \cdot k / \omega$:

Eq.2:

$$ED = k \cdot u_0^2 \cdot \eta \cdot \pi$$

In contrast, the energy dissipated in viscous damping increases linearly with the forcing frequency, as shown in Fig. 1.



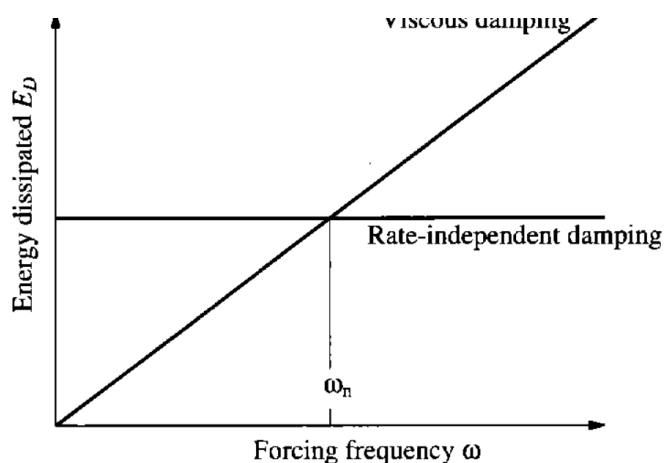


Fig. 1 - Viscous and Rate Independent Damping

Rate-independent damping is easily described if the excitation is harmonic and we are interested only in the steady-state response of this system. Difficulties arise in translating this damping mechanism back to the time domain. Thus it is most useful in the frequency-domain method of analysis (Section 1.10.3). The complex term $k(1 + i\eta)u$ represents both the elastic and damping forces at the same time; $k(1 + i\eta)$ is the complex stiffness of the system. There are advantages in this terminology if the reader is familiar with complex-variable notation. The complex stiffness has no physical meaning, however, in the same engineering sense as the elastic stiffness.

0.5-2 | Steady-State Response to Harmonic Force

The equation for an SDF system with rate-independent linear damping, denoted by a crossed box in Fig. 3.10.2, is Eq. (3.2.1) with the damping term replaced by Eq. (3.10.1):

Eq.3:

$$k \cdot u + \frac{k \cdot u \cdot \eta}{\omega} + m \cdot \ddot{u} = p(t)$$

The mathematical solution of this equation is quite complex for arbitrary $p(t)$. Here we consider only the steady-state motion due to a sinusoidal forcing function, $p(t) = p_0 \sin(\omega t)$, which is described by

Eq.4:

$$u(t) = u_0 \cdot \sin(t \cdot \omega - \varphi)$$

The amplitude u_0 and phase angle φ are

Eq.5:

$$u_0 = u_{st} \cdot 1 \cdot \frac{1}{\sqrt{\eta^2 + \left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right)^2}}$$

Eq.6:

$$\varphi = \arctan\left(\frac{\eta}{1 - \left(\frac{\omega}{\omega_n}\right)^2}\right)$$

These results are obtained by modifying the viscous damping ratio in Eqs. (3.2.11) and (3.2.12) to reflect the damping force associated with rate-independent damping, Eq. (3.10.1). In particular, ζ was replaced by

Eq.7:



$$\zeta = \frac{c}{c_c}$$

Eq.8:

$$\zeta = \frac{k \cdot \eta}{\omega \cdot 2 \cdot m \cdot \omega_n}$$

Eq.9:

$$\zeta = \frac{\eta}{\left(\frac{2 \cdot \omega}{\omega_n} \right)}$$

Shown in Fig 3 by solid lines are plots of $u_0/(u_{st})_0$ and ϕ as a function of the frequency ratio ω/ω_n for damping coefficient $\eta = 0, 0.2$, and 0.4 ; the dashed lines are described in the next section. Comparing these results with those in Fig. 3.2.6 for viscous damping, two differences are apparent: First, resonance (maximum amplitude) occurs at $\omega = \omega_n$, not at $\omega < \omega_n$. Second, the phase angle for $\omega = 0$ is $\phi = \arctan(\eta)$ instead of zero for viscous damping; this implies that motion with rate-independent damping can never be in phase with the forcing function.

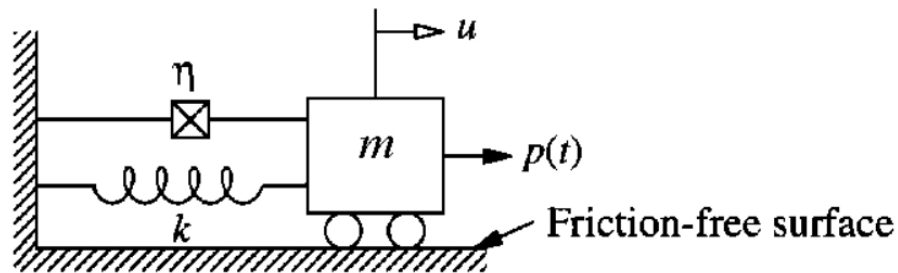


Fig. 2 - SDF Rate Independent Linear Damping

These differences between forced vibration with rate-independent damping and forced vibration with viscous damping are not significant, but they are the source of some difficulty in reconciling physical data. In most damped vibration, damping is not viscous, and to assume that it is without knowing its real physical characteristics is an assumption of some error. In the next section this error is shown to be small when the real damping is rate independent.



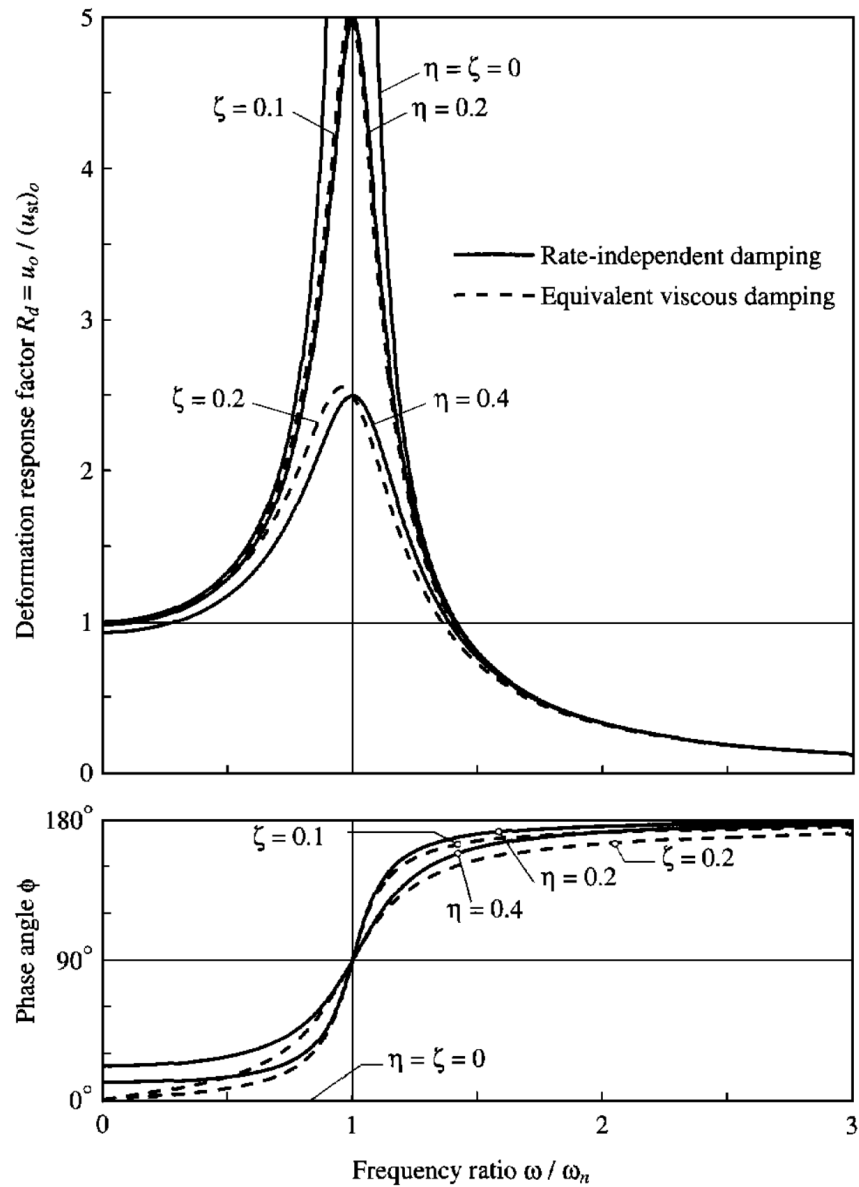


Fig. 3 - Rate Independent Damping

0.5-3 | Solution Using Equivalent Viscous Damping

In this section an approximate solution for the steady-state harmonic response of a system with rate-independent damping is obtained by modeling this damping mechanism as equivalent viscous damping.

Matching dissipated energies at $\omega = \omega_n$ (Fig. 3.10.1) led to Eq. (3.9.2), where ED is given by Eq. (3.10.2), leading to the equivalent viscous damping ratio:

Eq.10:

$$\zeta_{eq} = \frac{\eta}{2}$$

Substituting this ζ_{eq} for ζ in Eqs. (3.2.10) to (3.2.12) gives the system response. The resulting amplitude u_0 and phase angle ϕ are shown by the dashed lines in Fig. 3.10.3. This approximate solution matches the exact result at $\omega = \omega_n$ because that was the criterion used in selecting ζ_{eq} . Over a wide range of excitation frequencies the approximate solution is seen to be accurate enough for many engineering applications. Thus Eq. (3.10.3) - which is difficult to solve for arbitrary force $p(t)$ that contains many harmonic components of different frequencies ω - can be replaced by the simpler Eq. (3.2.1) for a system with equivalent viscous damping defined by Eq. (3.10.8). This is the basic advantage of equivalent viscous damping.



0.5-4 | Harmonic Vibration with Coulomb Friction

Equation of Motion

Shown in Fig. 3.11.1 is a mass-spring system with Coulomb friction force $F = \mu \cdot N$ that opposes sliding of the mass. As defined in Section 2.4, the coefficients of static and kinetic friction are assumed to be equal to μ , and N is the normal force across the sliding surfaces. The equation of motion is obtained by including the exciting force in Eqs. (2.4.1) and (2.4.2) governing the free vibration of the system:

$$m \cdot \ddot{u} + k \cdot u \pm F = p(t)$$

The sign of the friction force changes with the direction of motion; the positive sign applies if the motion is from left to right ($\dot{u} > 0$) and the negative sign is for motion from right to left ($\dot{u} < 0$). Each of the two differential equations is linear, but the overall problem is nonlinear because the governing equation changes every half-cycle of motion. Therefore, exact analytical solutions would not be possible except in special cases.

Steady-State Response to Harmonic Force

An exact analytical solution for the steady-state response of the system of Fig. 4 subjected to harmonic force was developed by J. P. Den Hartog in 1933.

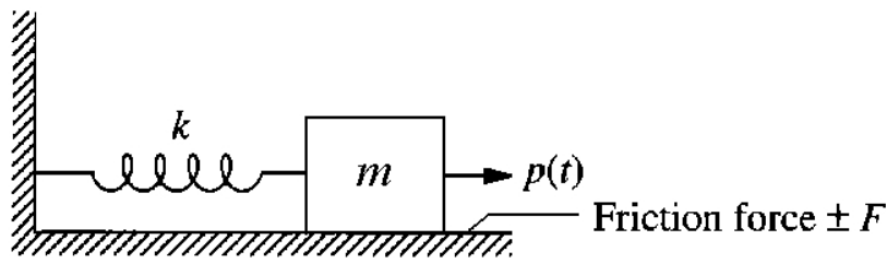


Fig. 4 - SDF Coulomb Damping

The analysis is not included here, but his results are shown by solid lines in Fig. 5. The displacement amplitude u_0 , normalized relative to $(u_{st})_0 = p_0/k$, and the phase angle ϕ are plotted as a function of the frequency ratio ω/ω_n for three values of F/p_0 . If there is no friction, $F = 0$ and $u_0/(u_{st})_0 = (Rd)_{\zeta=0}$, the same as in Eq. (3.1.11) for an undamped system. The friction force reduces the displacement amplitude u_0 , with the reduction depending on the frequency ratio ω/ω_n .

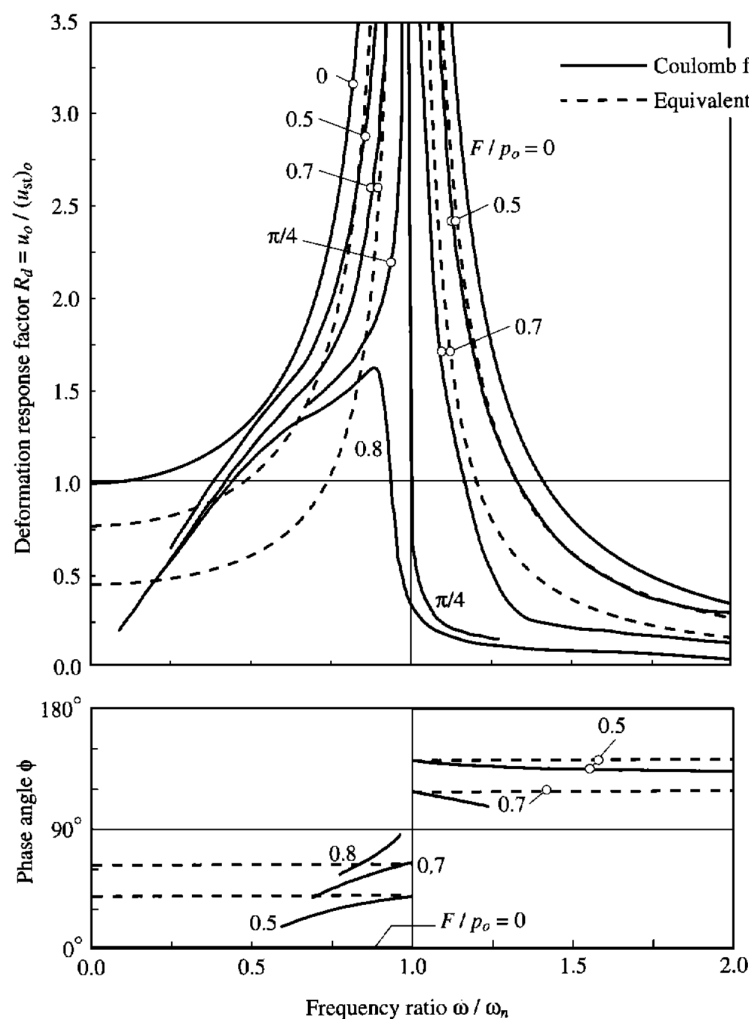


Fig. 5 - Damping with Coulomb Friction

Deformation response factor and phase angle of a system with Coulomb friction excited by harmonic force. Exact solution from J. P. Den Hartog; approximate solution is based on equivalent viscous damping.

[0.5.1]

Anil K. Anil K. Chopra, Dynamics of Structures: Theory and Applications to Earthquake Engineering. Englewood Cliffs, NJ, USA: Prentice Hall, 1995.

0.6-1 | Eigenvalues and Vectors

Analyze a 3-story shear frame using the flexibility method to determine natural frequencies and mode shapes (after Clough and Penzien [0.6.1])

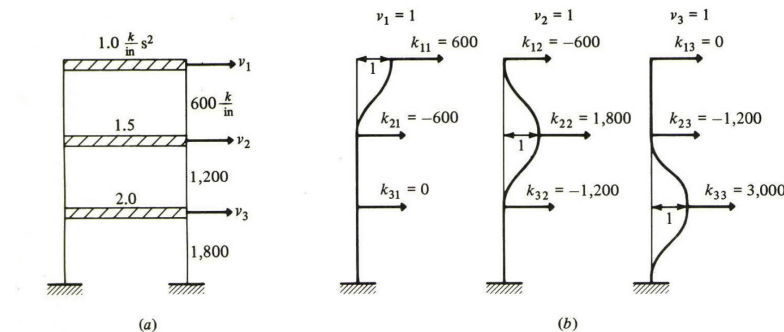


Fig. 1 - Structural Model

```
import numpy as np
import numpy.linalg as la
import textwrap as tw
# set up mass and stiffness arrays
m = np.array([[1.0,0,0],[0,1.5,0],[0,0,2.0]],float)
k1 = 600*np.array([[1,-1,0],[-1,3,-2],[0,-2,5]],float)
# flexibility and dynamic matrix
f = la.inv(k1)
d = np.inner(f,m)
eigen = la.eig(d)
evalus = eigen[0]
print("\neigenvalues:\n", " "*8,evalus)

eigenvalues:
      [0.00474206 0.00103739 0.00047055]
#
nat_freq = 1/(np.sqrt(evalus))
print("\nnatural frequencies:\n", " "*8,nat_freq)

natural frequencies:
      [14.52166783 31.04769646 46.09947622]
#
evect = np.array(eigen[1])
print("\neigenvectors:\n",tw.indent(str(evect), " "*8))

eigenvectors:
      [[-0.81332769 -0.73942881  0.27304451]
       [-0.52747169  0.44853685 -0.69406171]
       [-0.24550292  0.50205551  0.66612689]]
```

0.6-2 | Plot Mode Shapes

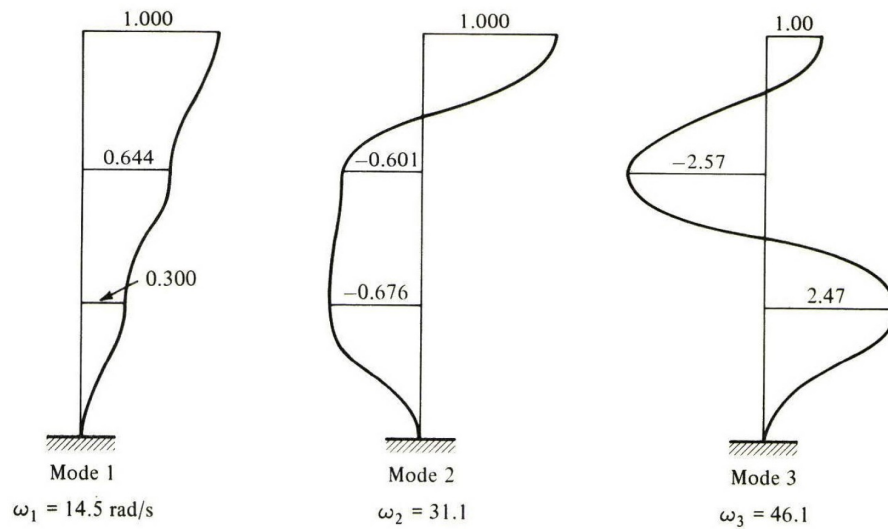


Fig. 2 - Structural Model

Plot normalized mode shapes and compare to Penzien and Clough. [0.6.2]

```
import numpy as np
import matplotlib.pyplot as plt
import os
from tabulate import tabulate as tb
# initialize eigenvector array (need (x,1) shapes for plotting)
ms = np.shape(evect)
zz = np.zeros((ms[0],1))
x1=np.concatenate((evect,zz),1)
#plot mode shapes using matplotlib
y=np.array([0,1,2,3])
m3=x1[2]*.35+5
m2=x1[1]*.35+3
m1=x1[0]*.35+1
m=np.concatenate((m1,m2,m3))
plt.clf()
plt.plot(m1,y)
plt.plot(m2,y)
plt.plot(m3,y)
plt.xlim(.5,6.)
plt.xlabel('mode')
plt.ylabel('levels')
plt.title("Mode Shapes")
plt.grid()
curdir=os.getcwd()
imgdir=os.path.join(curdir,"img","mode_shapes.png")
plt.savefig(imgdir)
# table of eigenvalues and normalized eigenvectors
evectt = np.transpose(evect)
for i in range(len(nat_freq)):
    evectt[i] = evectt[i]/evectt[i][0]
```

```
xx = np.concatenate((nat_freq[:, np.newaxis],evec),1)
xx = np.round(xx, 4)
yy = ["freq","level 3","level 2","level 1"]
tt = np.vstack((yy,xx))
print("\nTable of eigenvalues and normalized eigenvectors\n",
tb(tt, headers="firstrow", tablefmt="rst"))
```

Table of eigenvalues and normalized eigenvectors

freq	level 3	level 2	level 1
14.5217	1	1	1
31.0477	0.6485	-0.6066	-2.5419
46.0995	0.3018	-0.679	2.4396

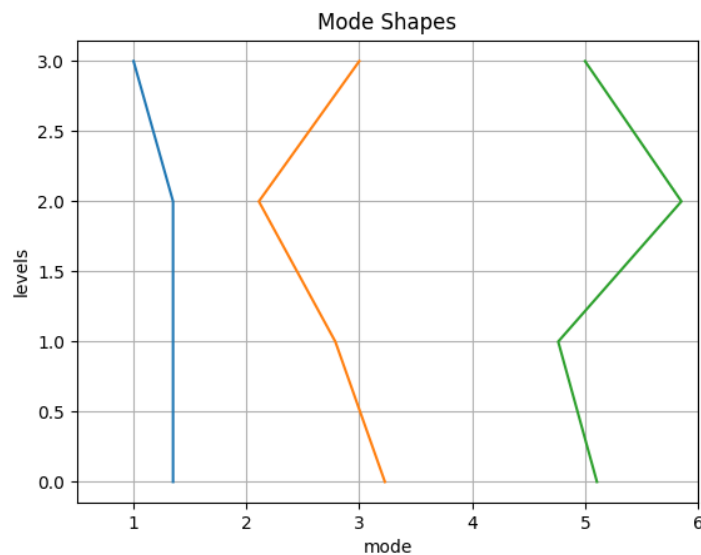


Fig. 3 - Calculated Normalized Modes

[0.6.1]

R.W. Clough and J. Penzien, Dynamics of Structures. New York, NY, USA:McGraw-Hill, 1975. pg. 178-180

[0.6.2]

ibid. pg. 180-182

